ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** burning of coal, ripening of fruit	Both produce new substances.
2	**(a)** no moisture	Moisture is needed.
3	**(d)** crystals form when a hot saturated solution cools	Cooling.
4	**(d)** freezing of water, evaporation of water	States change only.
5	**(d)** no air reaches it	Oil stops oxygen.
6	**(d)** to remove insoluble impurities	Clear solution.
7	**(c)** frying of an egg	Chemical change.
8	**(a)** both air and water are present	Conditions fulfilled.
9	**(d)** slow cooling gives bigger crystals	Time for growth.
10	**(b)** dissolving is a physical change	Original substance recovered.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Melting of wax is physical (1); Burning of wax vapour is chemical (1).	1+1
12	Solubility decreases with temperature fall (1); Extra solute separates (1).	1+1
13	Zinc coating protects (1); Zinc corrodes first (1).	1+1
14	New substance, iron oxide, is formed (1); It cannot be reversed (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nail turns brown (1); Solution turns pale green (1); Chemical change (1).	1+1+1
16	Plants make glucose (1); Uses carbon dioxide and water (1); New substances formed (1).	1+1+1
17	Cooking: chemical (1); Melting: physical (1); Irreversible vs reversible (1).	1+1+1
	OR: Fizzing (1); Carbon dioxide (1); Chemical change (1).	1+1+1
18	Oxygen causes spoilage (1); Moisture causes spoilage (1); Keeps it fresh (1).	1+1+1
	OR: Oil and paint (1); Wash and dry (1); Cover after use (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Two beakers: tap water and salt water (1); Same nails and exposure (1); Observe for a week (1); Salt water nail rusts more (1); Salt speeds rusting (1).	1+1+1+1+1
	OR: Useful: cooking (1); Useful: making curd (1); Useful: photosynthesis (1); Harmful: rusting (1); Harmful: food spoiling (1).	1+1+1+1+1
20	Fuel burns: chemical (1); Engine heats: physical (1); Paint protects (1); Rust forms: chemical (1); Exhaust gases (1).	1+1+1+1+1
	OR: Apply lemon juice (1); Keep in water (1); Cover (1); Refrigerate (1); Reduce contact with air (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** salt in sea air speeds rusting	Salt accelerates.
2	**(a)** copper sulphate	Blue crystals.
3	**(a)** a chemical change	New substance formed.
4	**(b)** stainless steel spoon	Alloy.
5	**(a)** evaporation of water	Salt pans.
6	**(d)** a physical change occurs	Only size changes.
7	**(a)** to keep air and moisture away	Painting protects.
8	**(c)** a physical change	Dissolving.
9	**(a)** a chemical change occurs	Gas is evolved.
10	**(a)** prevent contact with moisture and air	Prevent rust.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tearing: physical (1); Burning: chemical (1).	1+1
12	Repeats after equal intervals (1); Pendulum swings or moon phases (1).	1+1
13	Sugar can be recovered after dissolving (1); Burning forms carbon and new gases (1).	1+1
14	New substances form (1); Colour and taste change (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	State changes: physical (1); No new substance (1); Chemical: new substance (1).	1+1+1
16	Evaporate part of water (1); Cool (1); Collect crystals (1).	1+1+1
17	Vapour condenses (1); Same substance (1); Can evaporate again (1).	1+1+1
	OR: Salt is conserved (1); Dissolving is physical (1); No new substance (1).	1+1+1
18	It can be frozen again (1); Same substance (1); No new product (1).	1+1+1
	OR: Melts (1); Turns brown (1); Becomes black carbon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heat sugar (1); It melts then turns black (1); New substance carbon (1); Candle wax melts and burns (1); Heating can start chemical change (1).	1+1+1+1+1
	OR: Mix baking soda and vinegar (1); Fizzing (1); Pass gas through lime water (1); It turns milky (1); Carbon dioxide (1).	1+1+1+1+1
20	Oxidation spoils food (1); Germs cause spoilage (1); Airtight containers (1); Refrigeration slows (1); Salt and sugar preserve (1).	1+1+1+1+1
	OR: Colour change by painting is physical (1); Heating copper sulphate is chemical (1); Mixing paints physical (1); Check new substance (1); Not always (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** a chemical change	It charrs.
2	**(b)** change in colour and gas formation	Clues.
3	**(d)** still remains protected	Zinc is more reactive.
4	**(c)** water of crystallisation is lost	Dehydration.
5	**(d)** a chemical change	Reaction with air.
6	**(a)** slow chemical change	It takes days.
7	**(c)** a reversible chemical change	Hydration.
8	**(c)** chemical and irreversible	New substance.
9	**(d)** a new substance, iron oxide, forms	New product.
10	**(d)** melting and freezing	State changes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Slow cooling allows crystals to grow (1); Quick cooling gives powder (1).	1+1
12	Oil keeps out moisture and air (1); Both are needed for rust (1).	1+1
13	New substances are formed (1); Curd cannot become milk (1).	1+1
14	Salt speeds up rusting (1); Moist air (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Burning forms new substances (1); Chemical change (1); Irreversible (1).	1+1+1
16	Burning (1); Gases and colours (1); Heat and light (1).	1+1+1
17	Reversible chemical change makes new substance (1); Example: blue copper sulphate heating and adding water (1); Physical does not (1).	1+1+1
	OR: White dazzling flame (1); White powder forms (1); Chemical change (1).	1+1+1
18	Combines with oxygen (1); Adds mass (1); Water also combines (1).	1+1+1
	OR: Rusts (1); Gets weakened (1); Leaks (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heating wax: melts, then burns (1); Heating sugar (1); Heating copper (1); Initial is physical (1); Later new substances (1).	1+1+1+1+1
	OR: Earth revolves (1); Moon revolves (1); Regular cycles (1); Seasons due to tilt (1); Predictable (1).	1+1+1+1+1
20	Prepare a saturated solution (1); Filter (1); Hang a seed crystal (1); Leave undisturbed (1); Crystal grows (1).	1+1+1+1+1
	OR: Water boils: physical (1); Steam builds pressure (1); Pulses soften (1); New substances form (1); Cooking is chemical (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND